

DEVESH DATWANI

Robotics Engineer - Perception Systems

Charlotte, NC, USA

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TECHNICAL SCOPE

Technology : Deep Learning, Computer Vision, IP Networks, Distributed Systems, Camera Sensors

Tools: Kubernetes, Docker, AWS, Linux, ffmpeg, MySQL, GitHub, CI/CD, Redis, Qdrant, Vector DB, ROS, Gazebo

Libraries and Frameworks: PyTorch, OpenCV, NumPy, Pandas, Django, FastAPI, SQL, ONNX, TensorRT, ZMQ, Redis

EXPERIENCE

FyveBy

Charlotte, USA

Senior Computer Vision Engineer

June 2026 - Present

- Architecting SkyView's production aircraft detection stack, a dual PTZ camera pipeline of object detection, OC-SORT 2D-tracking, zone-crossing state machines, and OCR deployed across 10 hangars & processing 52M frames/day.
- Building a tail-number recognition pipeline with SVTR & Real-ESRGAN super-resolution, fusing repeated reads via character-aligned consensus voting against live ADS-B rosters, raising read accuracy from 85% to 94%.
- Building a camera-to-LiDAR registration pipeline solving extrinsic via the 6 point algorithm with PnP and Rodrigues decomposition, achieving a base for sensor fusion for 3D-tracking with a Kalman filter.
- Fusing LiDAR & camera into a single 3D-tracker via voxel-down-sampled, coordinate-hashing, connected-component segmentation & projective cluster-to-bbox association, resolving aircraft identity across 2 cameras.
- Closing the data-flywheel with an in-production hard-sampler miner % and a custom ML platform for dataset curation, auto-labeling, training & evaluation.
- Shipping operator-level observability tools, including a web-socket based viewer with browser-based calibration, per-frame latency telemetry, Slack event sinks & recall dashboard for tracking daily detection accuracy across multiple sites.
- Owning dataset curation & governance across thousands of images, defining sampling strategy, class balance & label-schema versioning against deployed model ordering, & driving retraining & fine-tuning cycles that lifted mAP from 84% to 96%.

Bolt6

Charlotte, USA

Development and Operations Engineer

Sept 2023 - May 2026

- Owning maintenance and technical direction of a real-time object tracking system for a premier golf organization, featuring 120+ 4K cameras, cloud video pipelines, deep learning inference, and monitoring dashboards.
- Designing and deploying hybrid infrastructure, ranging from EKS clusters to on-site network architecture for RTSP-based ingestion of H.265 streams into AWS VPCs for real-time deep learning inference.
- Leading sensor fusion initiatives that combine radar and optical modalities to boost mean average precision of object tracking system from 45% to 96%, while maintaining sub-second latency and centimeter-level accuracy.
- Spearheading continuous 2D object tracking enhancements by analyzing and addressing class imbalance, curating and annotating datasets and iteratively fine-tuning YOLO-based object detection models for premier sporting events
- Maintaining GitHub Actions CI/CD pipelines to automate model validation, versioning and distribution with AWS stack, ensuring end-to-end reproducibility and reusability.
- Designing and deploying a multi-camera tracking pipeline with transformer-based ReID embeddings, centralized vector database for multi-camera matching, and ByteTrack-style association, achieving 92% ID accuracy.
- Building object masking pipelines enabling high-fidelity point cloud-image registration for accurate drift correction in drone localization.

LAB EXPERIENCE

Worcester Fire Department and WPI

Worcester, USA

Graduate Student Researcher

Aug 2022 - Dec 2022

- Developed a fire safety evaluation system based on robovacs in consultation with the Worcester Fire Department.
- Integrated software stack for SLAM in indoor environments with LDS-01 LIDAR, ICP registration and particle filters.
- Implemented a change detection algorithm with PCA and K-Means clustering to localize objects on lidar maps.
- Simulated the system in ROS and Gazebo environment to demonstrate applications to 350,000+ yearly fires in the USA.

Popovic Labs

Worcester, USA

Graduate Student Researcher

Aug 2021 - Dec 2021

- Represented Worcester Polytechnic Institute in a team of 15 students at the NASA Big Idea Challenge 2022.

- Designed sensor stack for a wheeled solution and wrote complementary filter in C++ for robust attitude estimation.

EDUCATION

Worcester Polytechnic Institute

Master of Science - Robotics Engineering, 3.5 GPA

Worcester, MA, USA

Aug 2021 - May 2023

RECOGNITION

Product development and operations lead for Emmy-winning (George Wensel Technical Award) Drone AR Project in 2025.